

Single-cell RNA sequencing data imputation using bi-level feature propagation

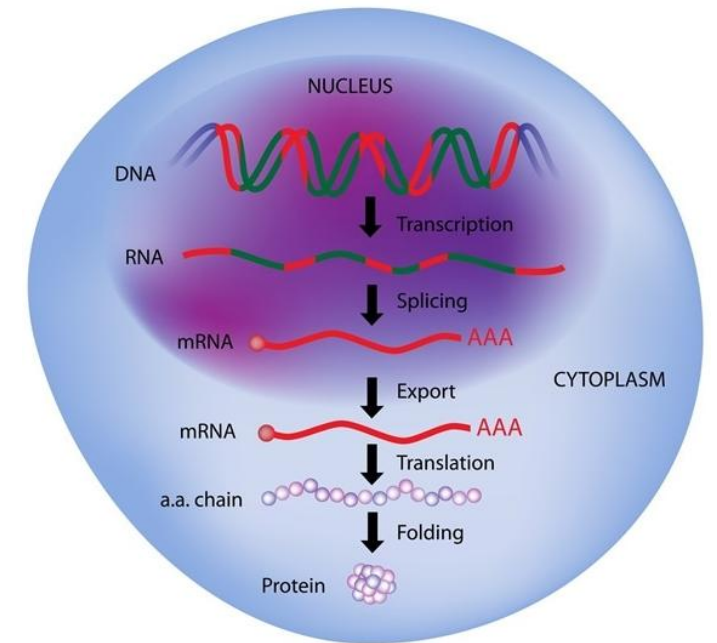
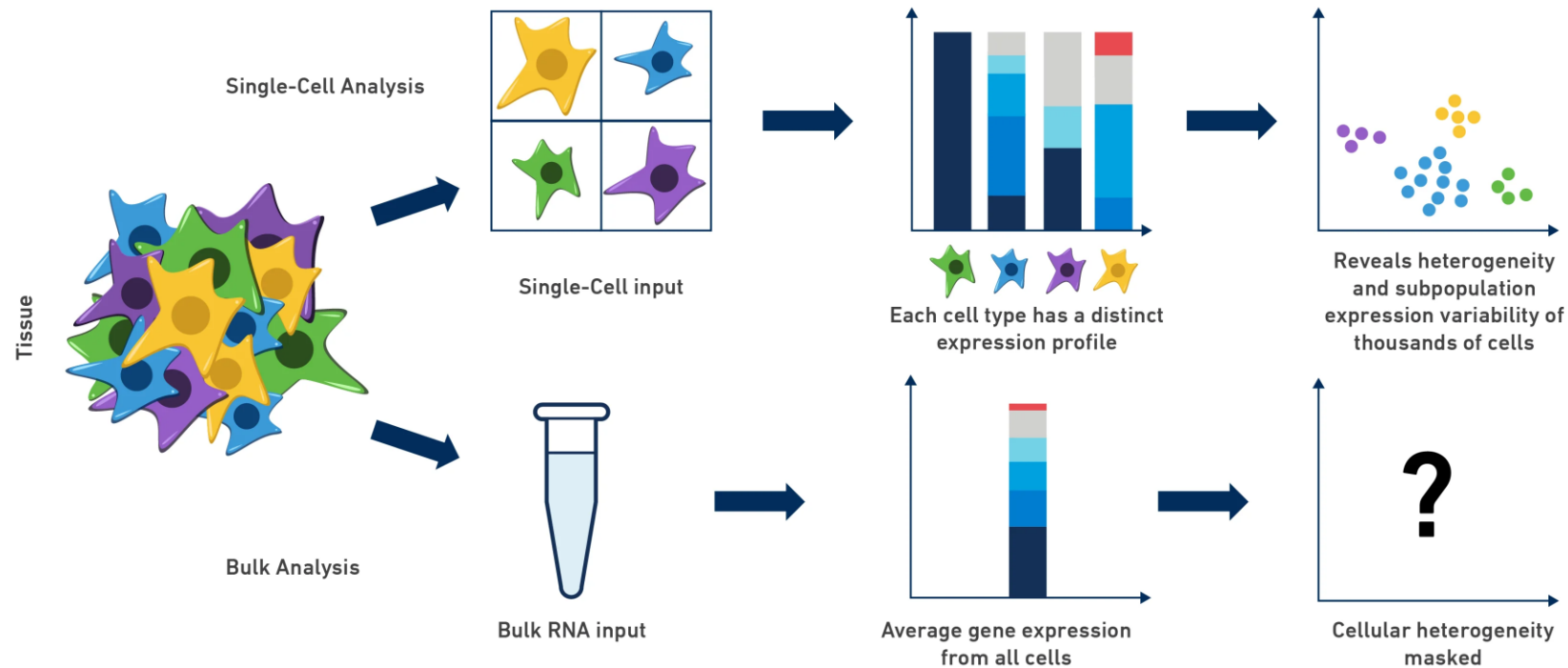
Briefings in Bioinformatics, 2024

WOOJIN LEE

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1. Preliminaries
 2. Background & Motivation
 3. Proposed Method
 4. Experiments & Results
 5. Conclusion & Discussion
 6. Further Analysis via Reproduction Experiments

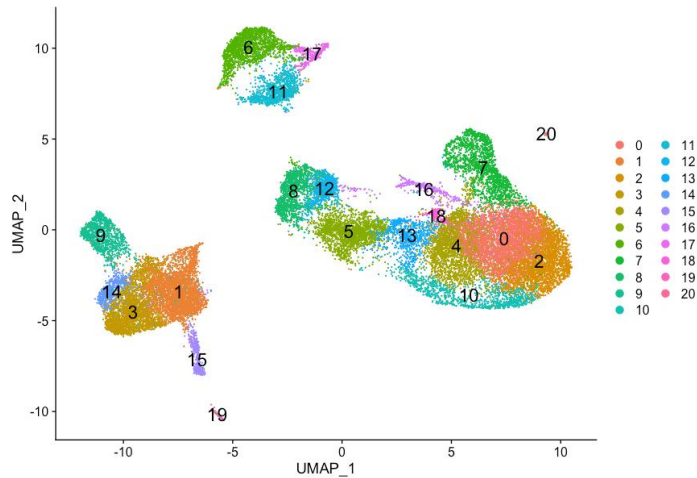
Transcriptomics, Single-Cell RNA Sequencing

Single-Cell RNA Sequencing: profiling gene expression at the individual cell level to reveal cellular heterogeneity

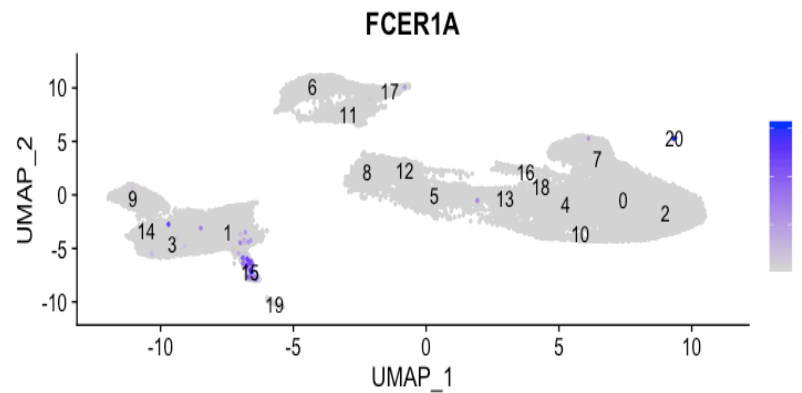


scRNA-seq Downstream Analysis

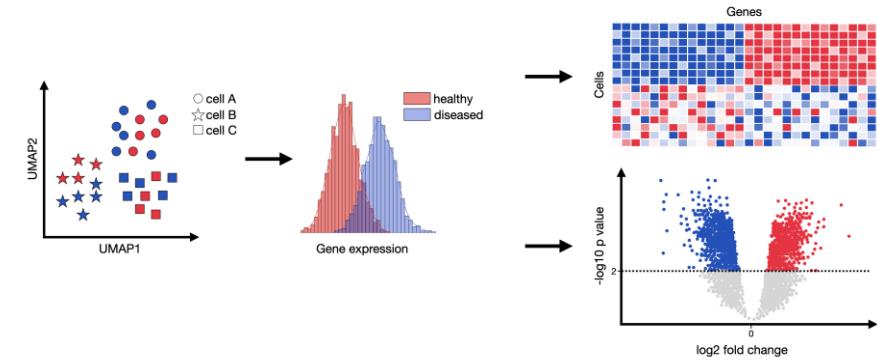
Cell Clustering



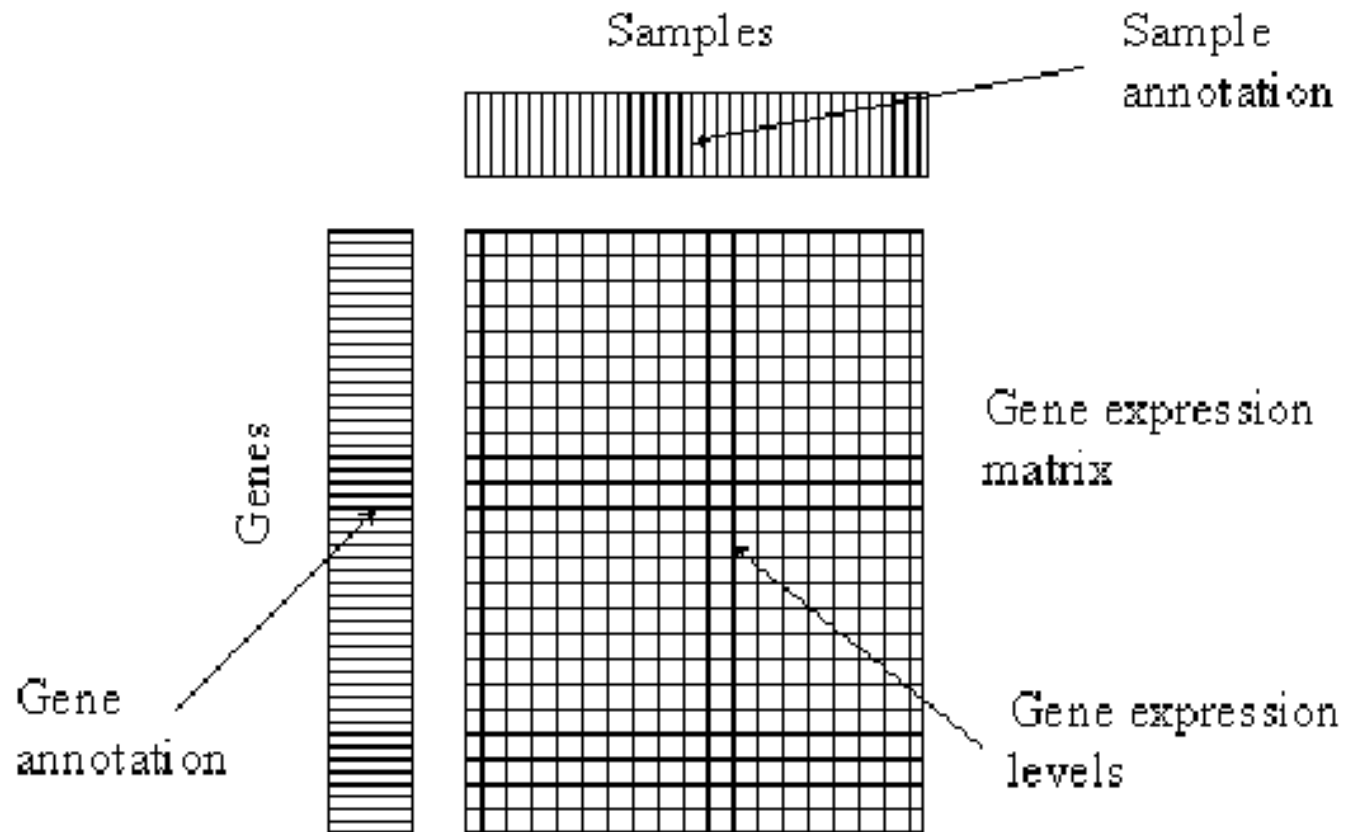
Marker Gene Identification



Differentially Expressed Gene (DEG) Analysis



Gene Expression, Gene-Cell Matrix



	Cell1	Cell2	...	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
...	.	.	.	.
...	.	.	.	.
...	.	.	.	.
GeneM	25	0	.	0

Feature Propagation

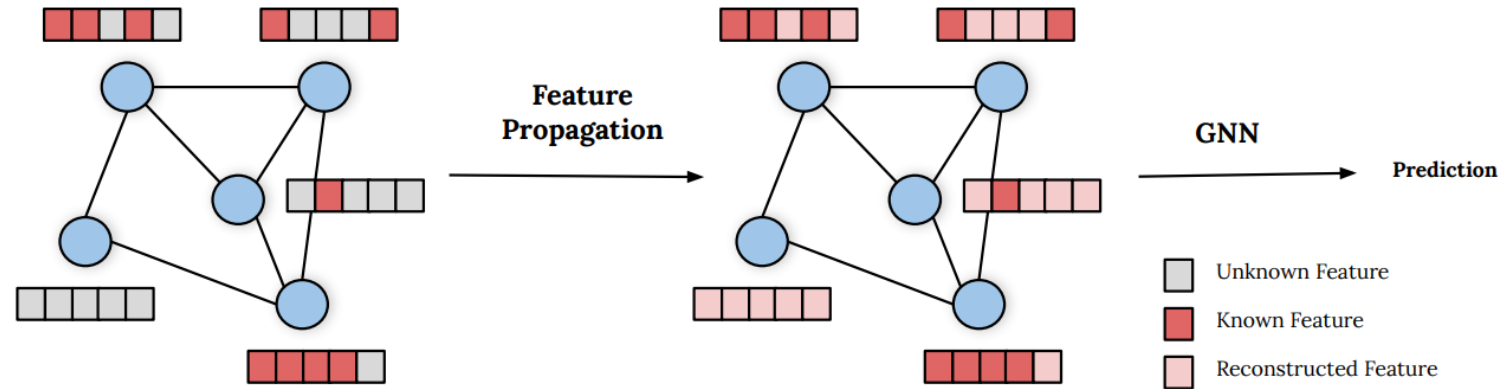
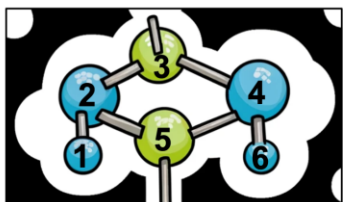


Figure 1: A diagram illustrating our Feature Propagation framework. On the left, a graph with missing node features. In the initial reconstruction step, Feature Propagation reconstructs the missing features by iteratively diffusing the known features in the graph. Subsequently, the graph and the reconstructed node features are fed into a downstream GNN model, which then produces a prediction.

Feature Propagation is a graph-based imputation method that fills missing features by diffusing observed information from neighboring nodes while keeping known values fixed.

Graph & Adjacency Matrix



Molecular graph
of six atoms

	f1	f2	f3	f4
1				
2				
3				
4				
5				
6				

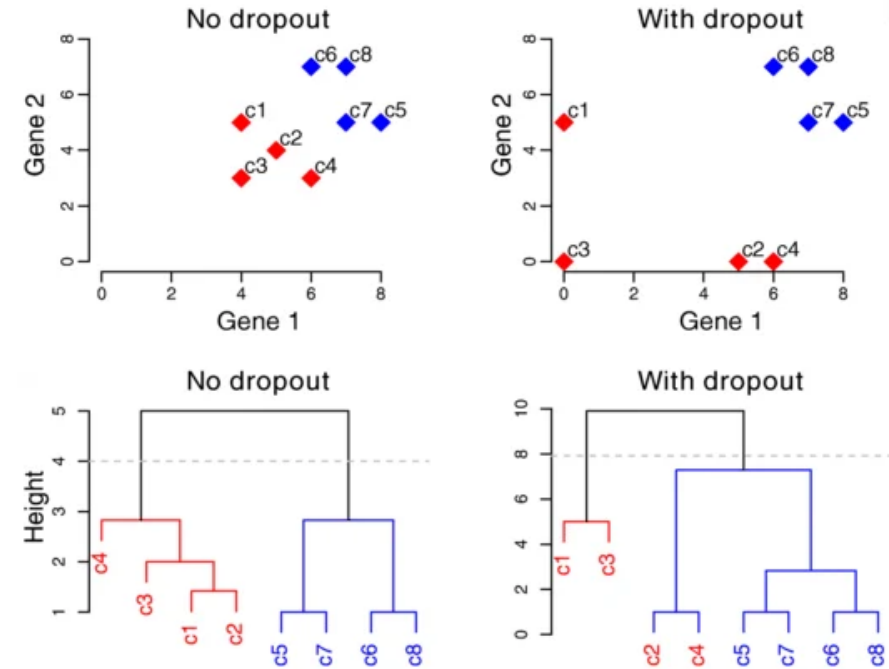
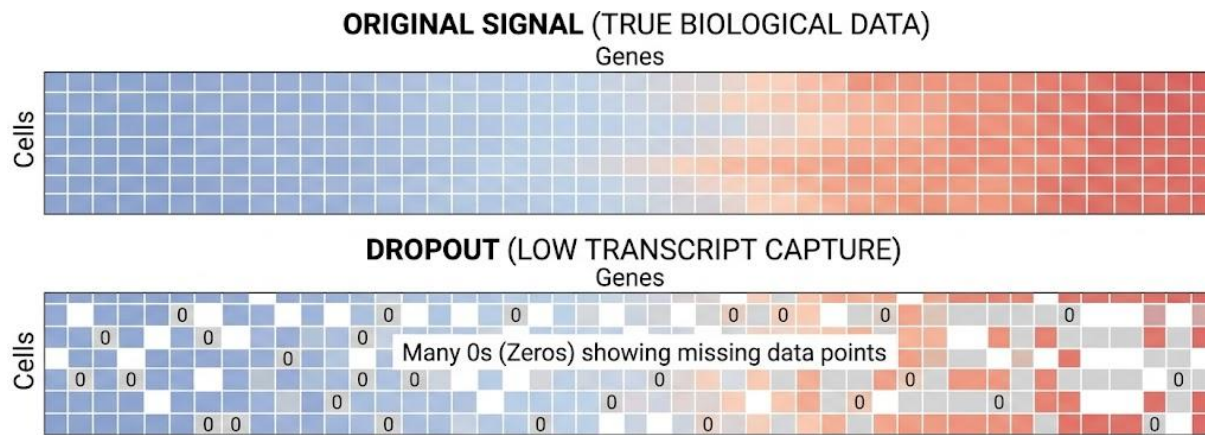
Node feature
matrix

	1	2	3	4	5	6
1	1	1	0	0	0	0
2	1	1	1	0	1	0
3	0	1	1	1	0	0
4	0	0	1	1	1	1
5	0	1	0	1	1	0
6	0	0	0	1	0	1

Adjacency
matrix

Labeled graph	Degree matrix
	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$
Adjacency matrix	Laplacian matrix
$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ -1 & 3 & -1 & 0 & -1 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & -1 & 0 & -1 & 3 & -1 \\ 0 & 0 & 0 & -1 & -1 & 2 \end{bmatrix}$

Technical Noise, Dropout Events, and Sparsity of scRNA-seq Data

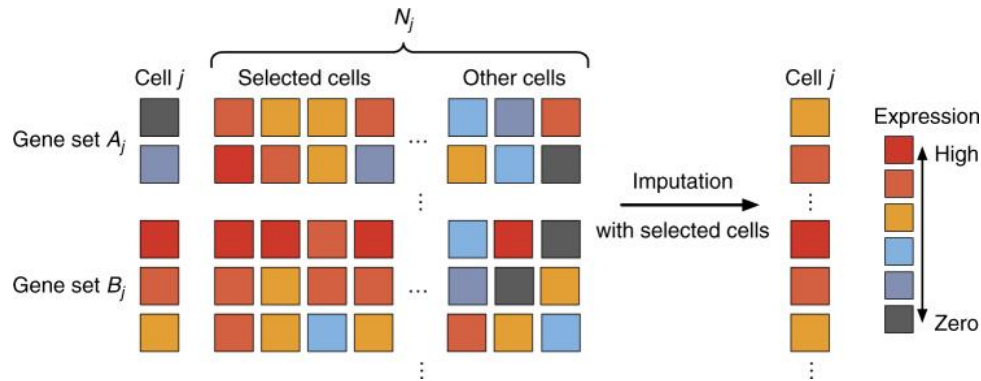


True Zero? or False Zero?
These complicate data preprocessing and hinder downstream analyses.

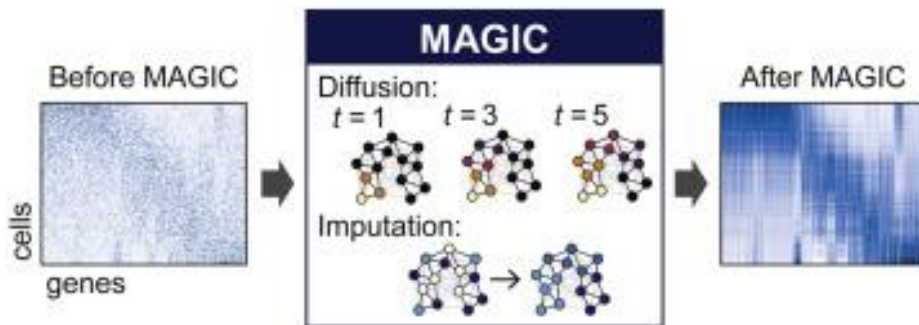
2 Background & Motivation

Previous Works

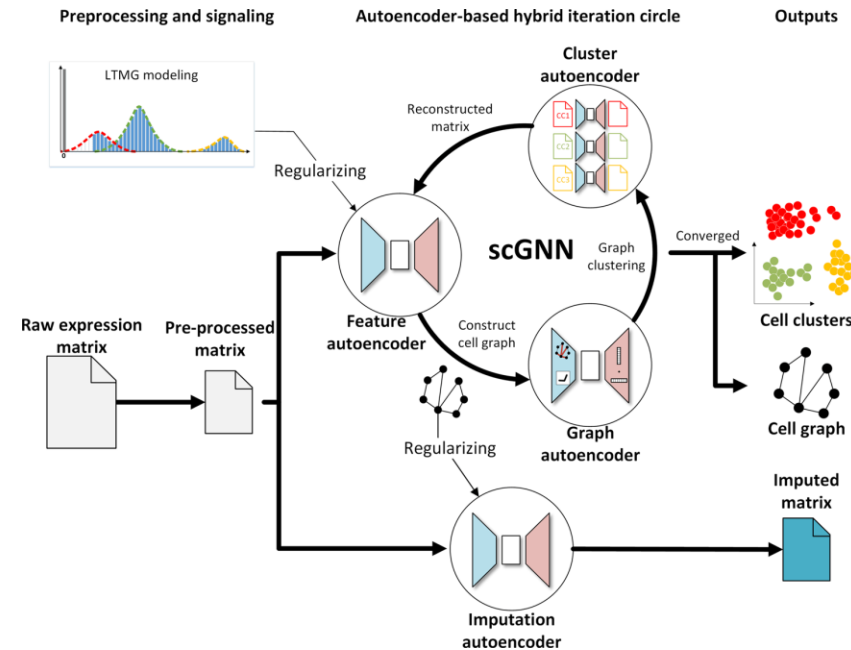
scImpute (non-graph-based)



MAGIC (graph-based)



scGNN (graph-based)



Non-graph-based

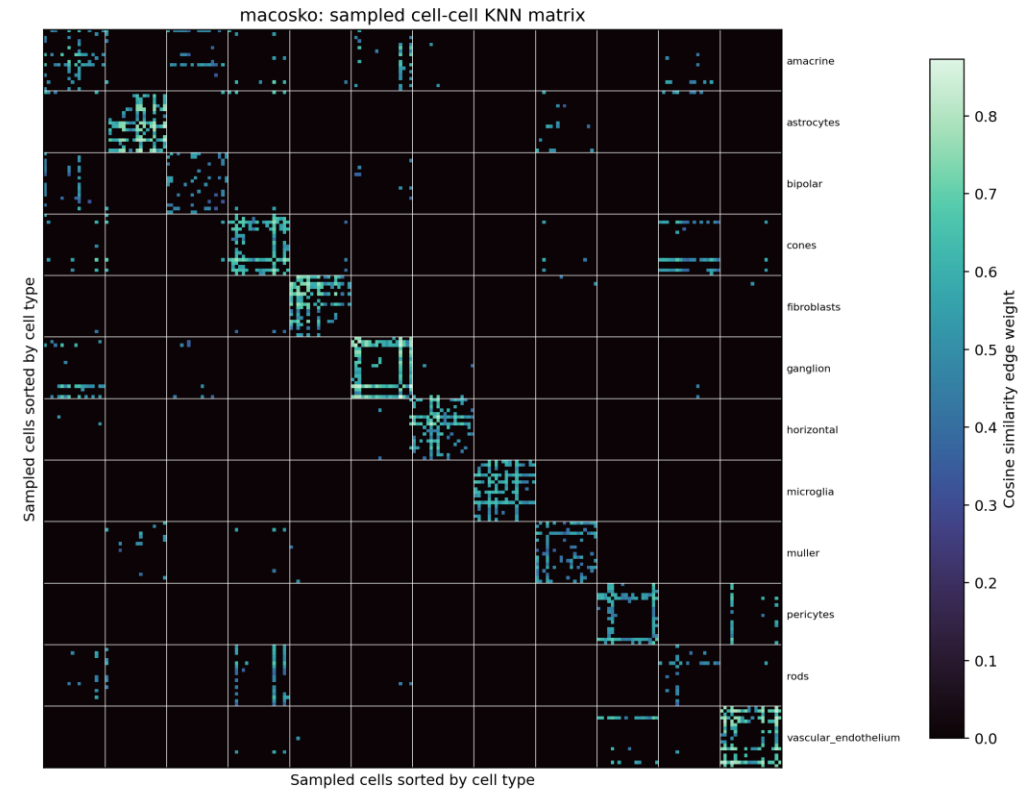
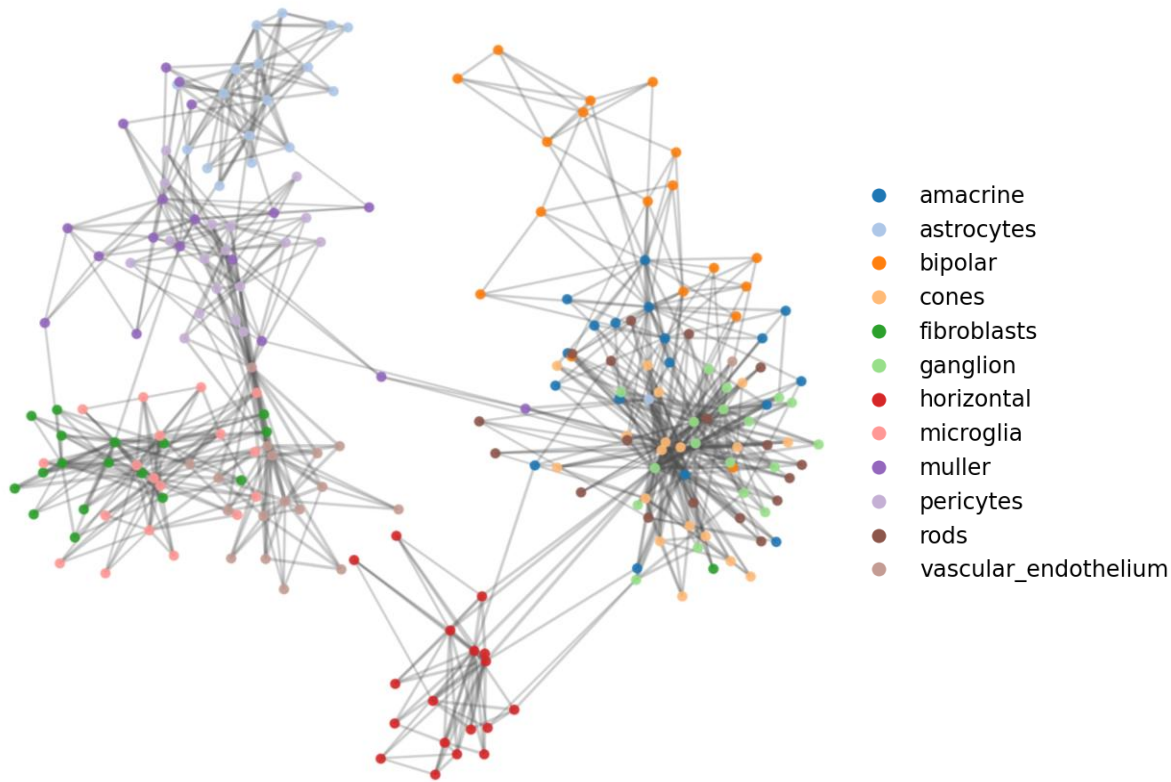
- not leveraging the relational structure among cells and genes

Graph-based

- vulnerable to noise propagation
- using initial graph from sparse/dropout-contaminated observations
- prioritizing either cell-cell or gene-gene relationships

Previous Works

Cell-Cell Graph Example & Adjacency-like Sample Matrix – via Macosko Dataset Sampling (knn k=4, no self-loop)



scBFP: single-cell Bi-level Feature Propagation Overview

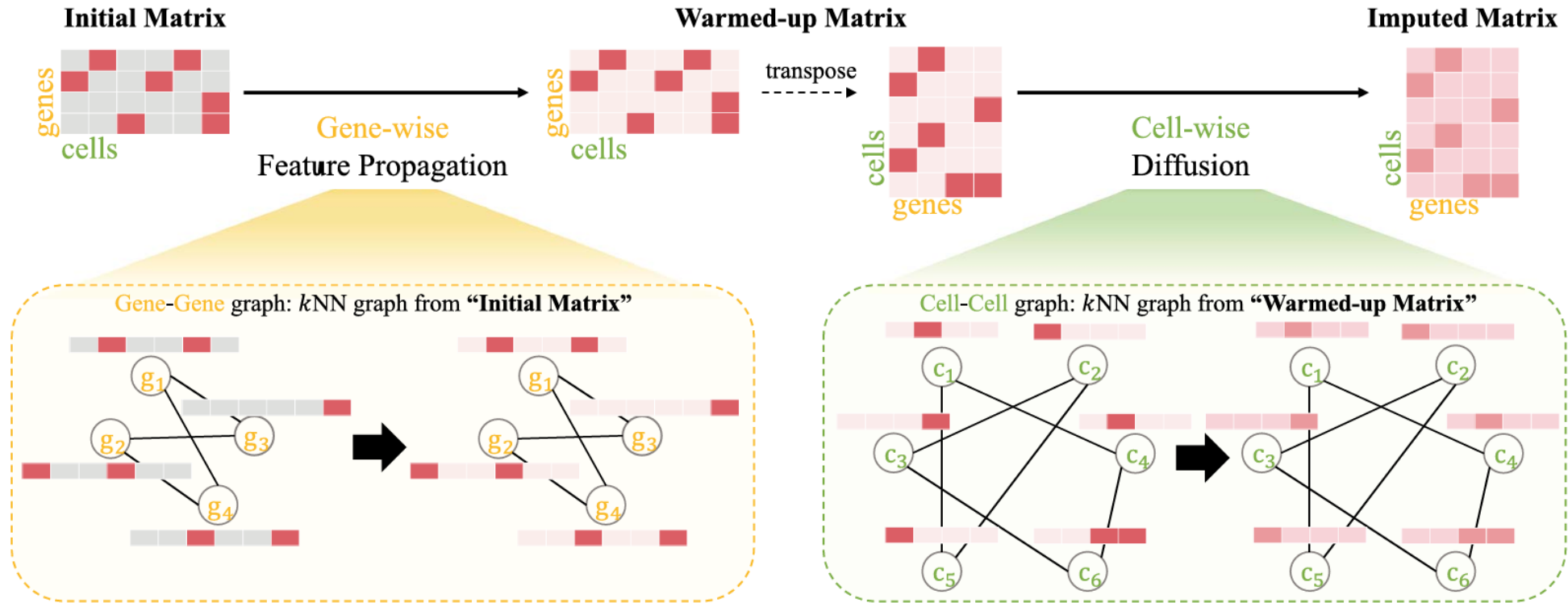


Figure 1. The overall framework of scBFP. Given an initial gene–cell count matrix, scBFP first conducts Gene-wise Feature Propagation on a gene–gene graph derived from this matrix, yielding a warmed-up matrix. Subsequently, using this warmed-up matrix, we derive an enhanced cell–cell graph and carry out diffusion to obtain the final imputed matrix.

scBFP: single-cell Bi-level Feature Propagation Overview

Feature Propagation Objective: Minimize the Dirichlet Energy

$$\ell(X, \mathcal{G}^{\text{gene}}) = \sum_{c=1}^C \frac{1}{2} X_{\cdot, c}^T \Delta X_{\cdot, c},$$

$$\Delta \in \mathbb{R}^{G \times G} = I - \tilde{A}^{\text{gene}}$$

$$X^{(i+1)} = \begin{bmatrix} I & 0 \\ \tilde{A}_{zn}^{\text{gene}} & \tilde{A}_{zz}^{\text{gene}} \end{bmatrix} X^{(i)}$$

Diffusion: Markov Chain

$$X'^{(j+1)} = \tilde{A}^{\text{cell}} X'^{(j)}$$

X : gene-cell matrix	X' : cell-gene matrix
$\mathcal{G}^{\text{gene}}$: gene-gene graph	D : degree matrix
$\mathcal{G}^{\text{cell}}$: cell-cell graph	Δ : graph Laplacian matrix
$\tilde{A}^{\text{gene}}$: normalized adjacency matrix based on gene-gene relationships	
$\tilde{A}^{\text{cell}}$: normalized adjacency matrix based on cell-cell relationships	

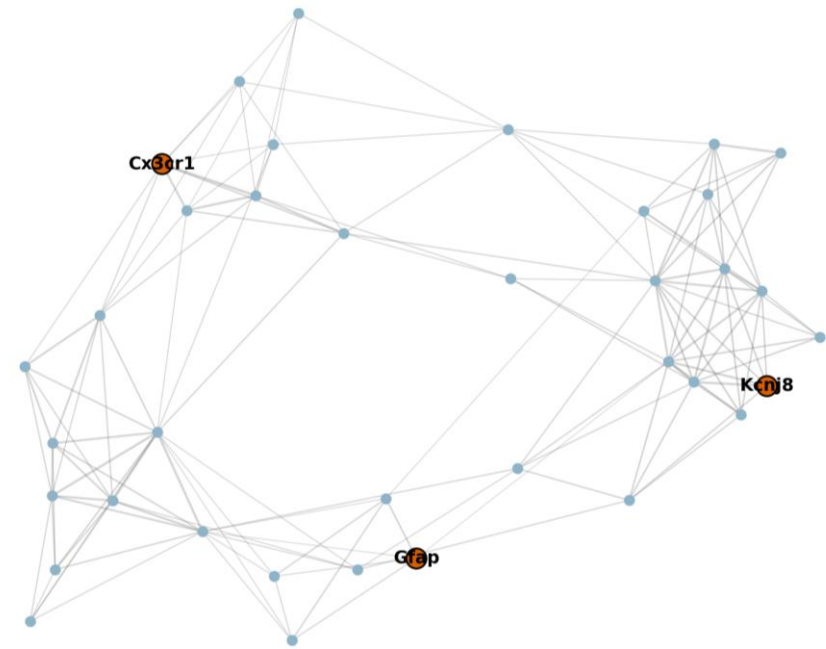
cf. The Convergence Property of scBFP

- **unknown-unknown propagation block's spectral radius < 1.** (FP Lemma A.1)
spectral radius = abs max eigenvalue
- **knn(k>10) graph empirically has strong connection.**
if. irreducible, finite state space
→ positive recurrent
→ unique stationary distribution = limiting distribution

scBFP: single-cell Bi-level Feature Propagation Algorithm

Algorithm 1 single-cell scBFP

```
1: Input: Gene-Cell Matrix  $X$ 
2: Output: Denoised Cell-Gene Matrix  $\hat{X}$ 
3:  $\tilde{A}^{\text{gene}} = \text{kNN}(X)$  ▷ Gene-Gene Graph Construction
4:  $Y \leftarrow X$ 
5: while  $X$  has not converged do
6:    $X \leftarrow \tilde{A}^{\text{gene}} X$ 
7:    $X_{\mathcal{N}} \leftarrow Y_{\mathcal{N}}$  ▷ Gene-wise Feature Propagation
8: end while
9:  $X' \leftarrow X^{(I)\top}$ 
10:  $\tilde{A}^{\text{cell}} = \text{kNN}(X')$  ▷ Cell-Cell Graph Construction
11: while  $X'$  has not converged do
12:    $X' \leftarrow \tilde{A}^{\text{cell}} X'$  ▷ Cell-wise Diffusion
13: end while
14:  $\hat{X} \leftarrow X'^{(I)}$ 
```



→ cosine-similarity-based knn gene-gene graph

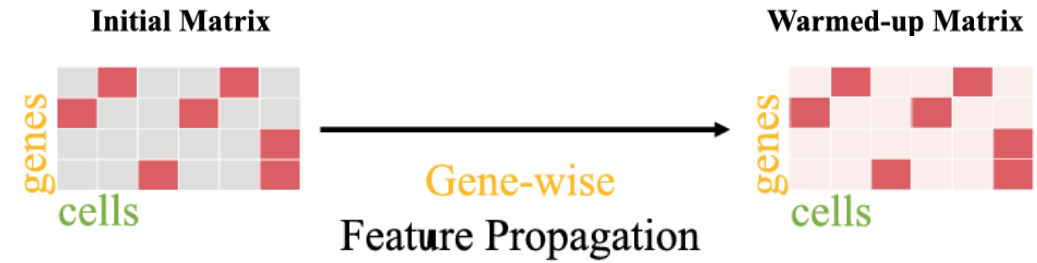
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12:    $X' \leftarrow \tilde{A}^{\text{cell}} X'$  ▷ Cell-wise Diffusion
13: end while
14:  $\hat{X} \leftarrow X'^{(I)}$ 

```



$$\tilde{A}^{\text{gene}} = D^{-1/2} A^{\text{gene}} D^{-1/2}$$

→ symmetrically normalized adjacency matrix

$$X^{(i+1)} = \begin{bmatrix} I & 0 \\ \tilde{A}_{zn}^{\text{gene}} & \tilde{A}_{zz}^{\text{gene}} \end{bmatrix} X^{(i)}$$

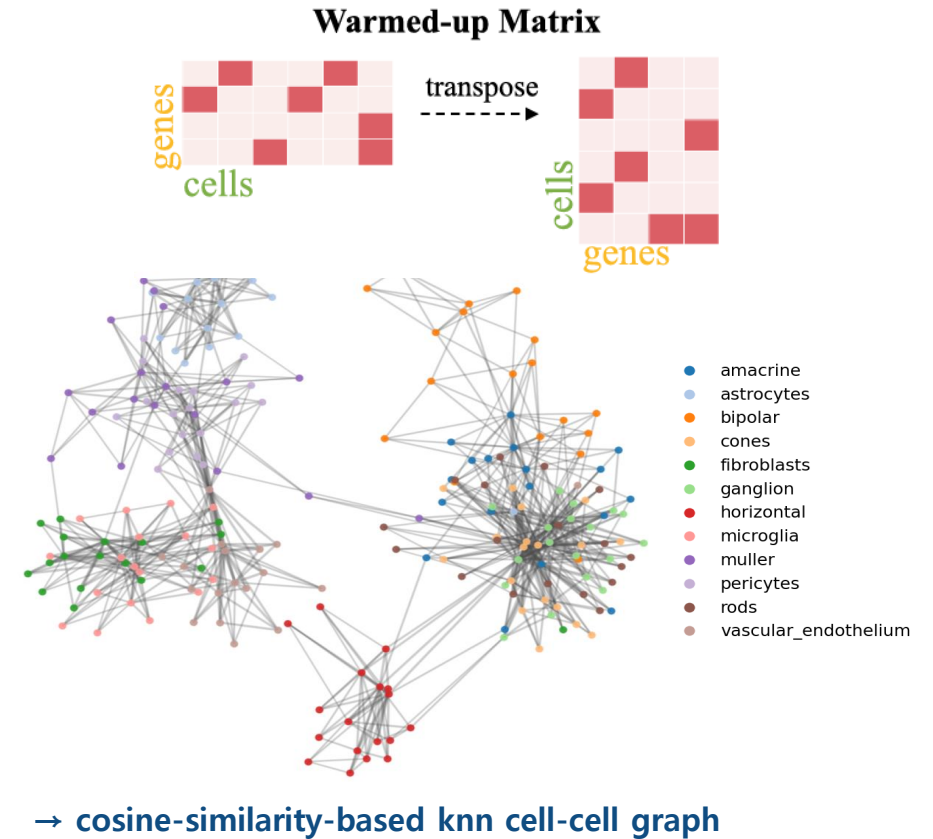
→ impute only zero values based on gene-gene relationships

3 Proposed Method

scBFP: single-cell Bi-level Feature Propagation Algorithm

Algorithm 1 single-cell scBFP

```
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2: Output: Denoised Cell-Gene Matrix  $\hat{X}$ 
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```



scBFP: single-cell Bi-level Feature Propagation Algorithm

Algorithm 1 single-cell scBFP

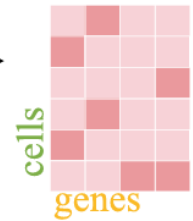
```
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```

Warmed-up Matrix



Cell-wise
Diffusion

Imputed Matrix



$$\tilde{A}^{\text{cell}} = D^{-1} A^{\text{cell}}$$

→ random-walk based normalization

$$X'^{(j+1)} = \tilde{A}^{\text{cell}} X'^{(j)}$$

→ not only impute zero values but also denoise non-zero values

4 Experiments & Results

Cell Clustering Performance compared to Baselines

Data	Sequencing platform	# of Cells	# of Genes	# of cell types
Baron mouse	inDrop	1,886	14,861	13
Pancreas	inDrop	1,937	15,575	14
Mouse ES cells	inDrop	2,717	24,047	4
Mouse bladder cells	Microwell-seq	2,746	19,771	16
Zeisel	STRT-seq UMI	3,005	19,972	7
Worm neuron cells	sci-RNA-seq	4,186	13,488	10
Baron Human	inDrop	8,569	17,499	14
Macosko mouse retina cells	Drop-seq	44,808	23,288	19

Supplementary Table S1: Statistics for scRNA-seq datasets with gold-standard cell types.

- generally improved clustering performance
- better than complex deep learning-based
- MAGIC also leverages relationship info
- Addressing GPU mem issue via feature-wise batch propagation

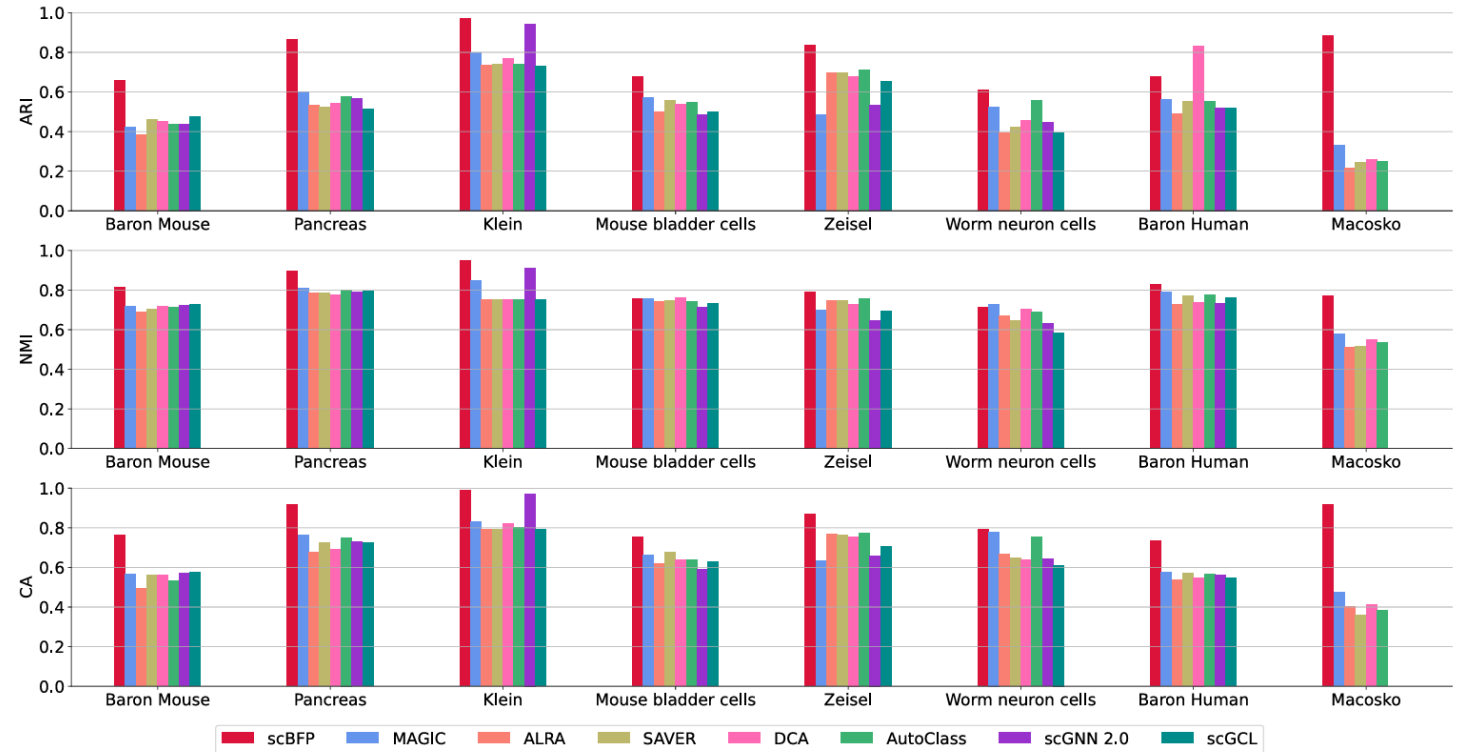
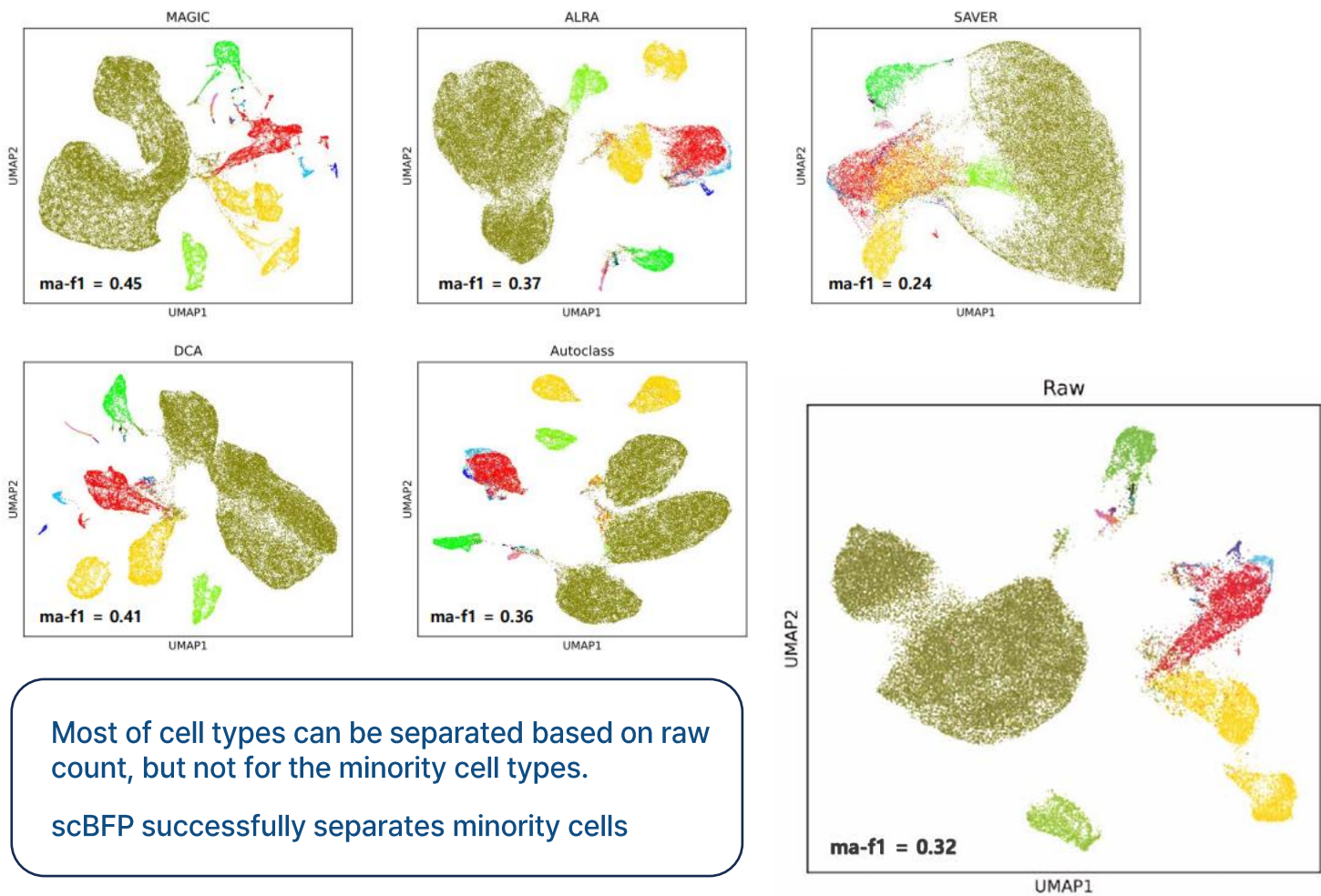


Figure 2. Performance comparisons of scBFP and other baselines on the eight scRNA-seq datasets.

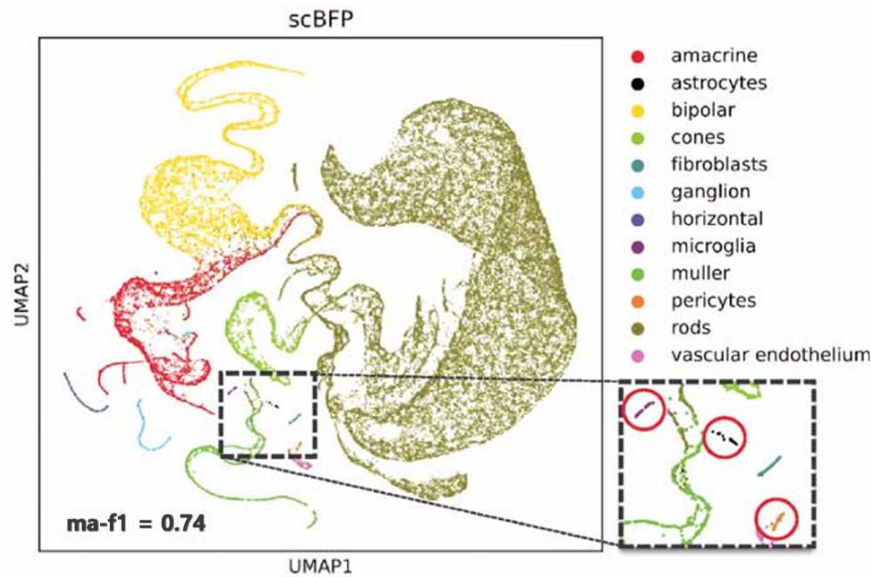
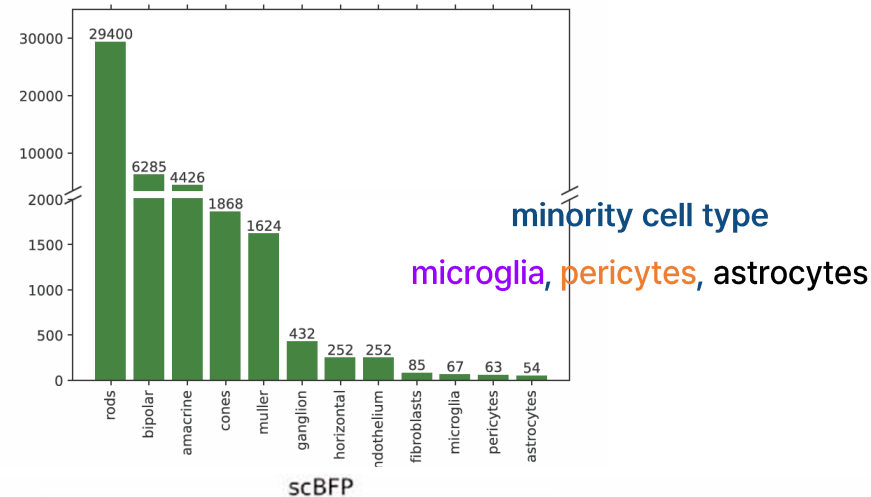
- ARI: pairwise clustering agreement adjusted for chance
- NMI: shared information between clusters and true labels
- CA: cell assignment accuracy after label matching

Rare Cell Type Identification via Macosko

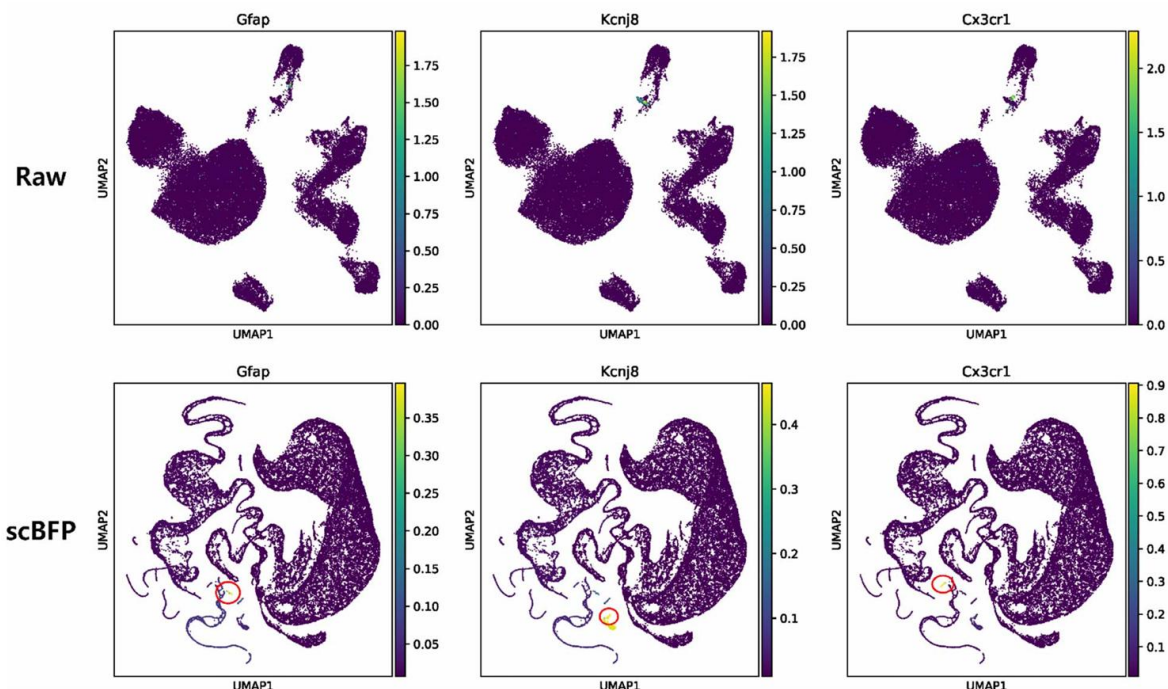


Most of cell types can be separated based on raw count, but not for the minority cell types.
scBFP successfully separates minority cells

Macosko Dataset Cell Type Distribution



Marker Gene Expression Identification

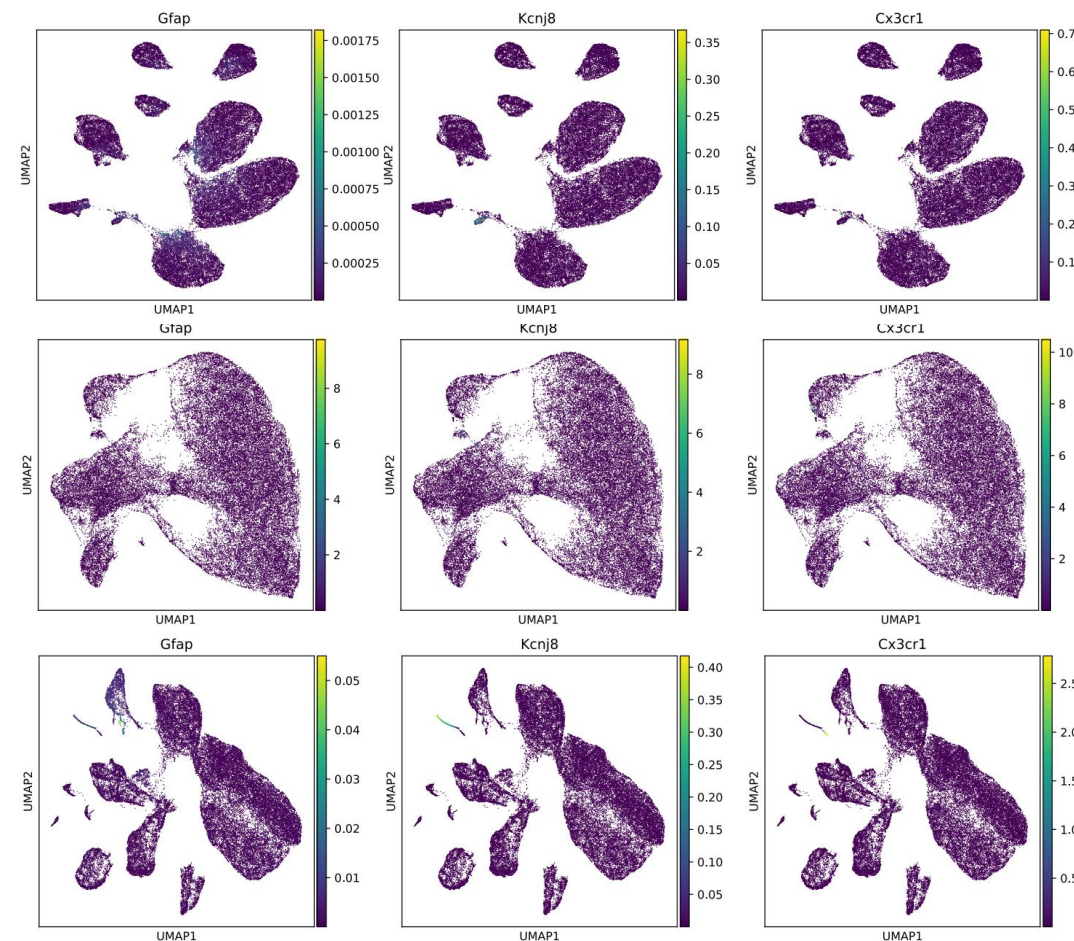


scBFP makes marker gene expression more cell-type-specific, while raw counts and baselines show less clear separation across cell types.

AutoClass

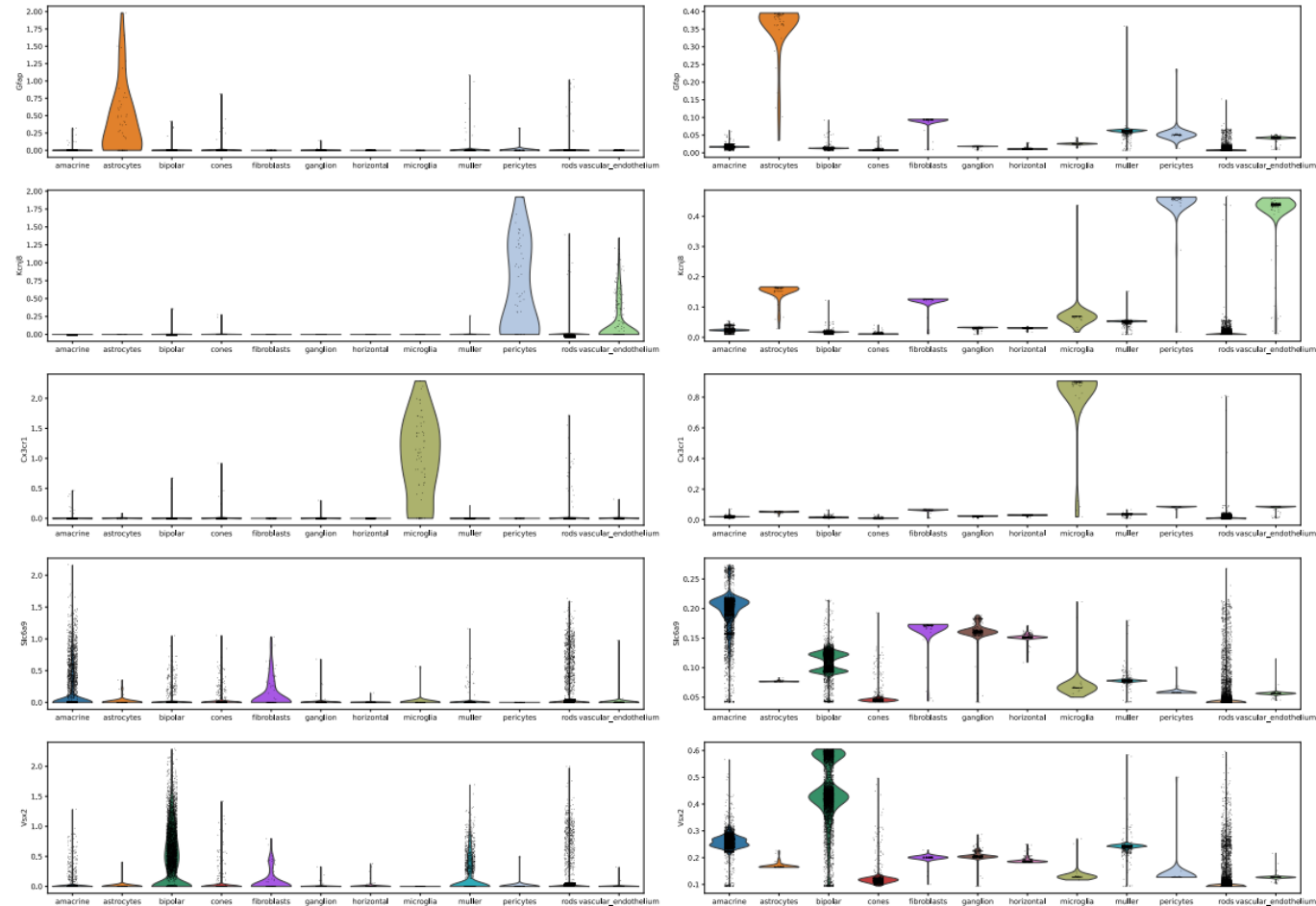
SAVER

DCA



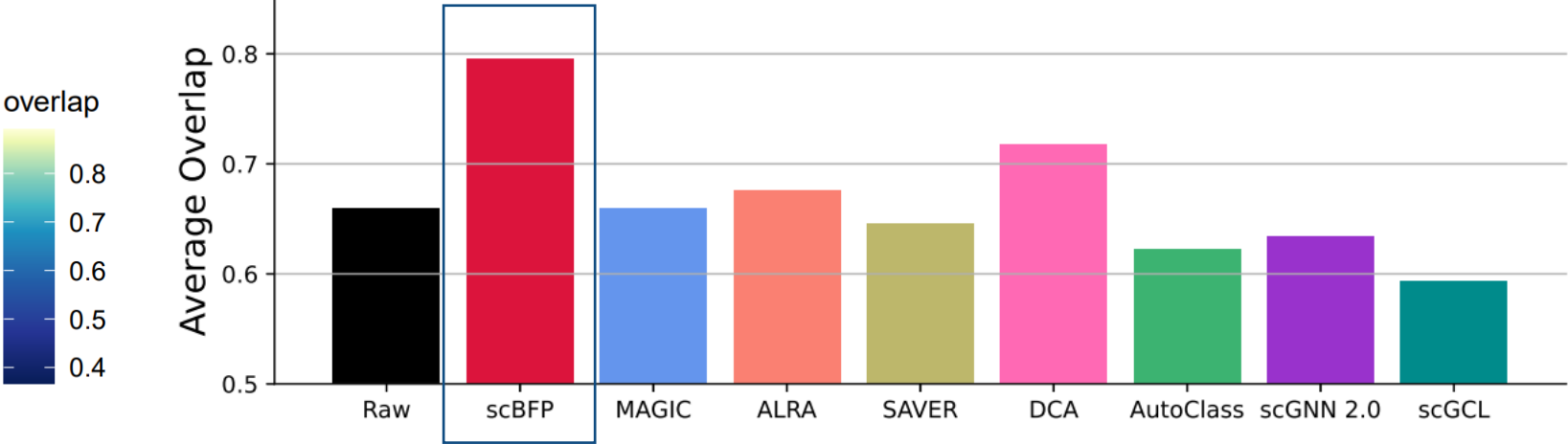
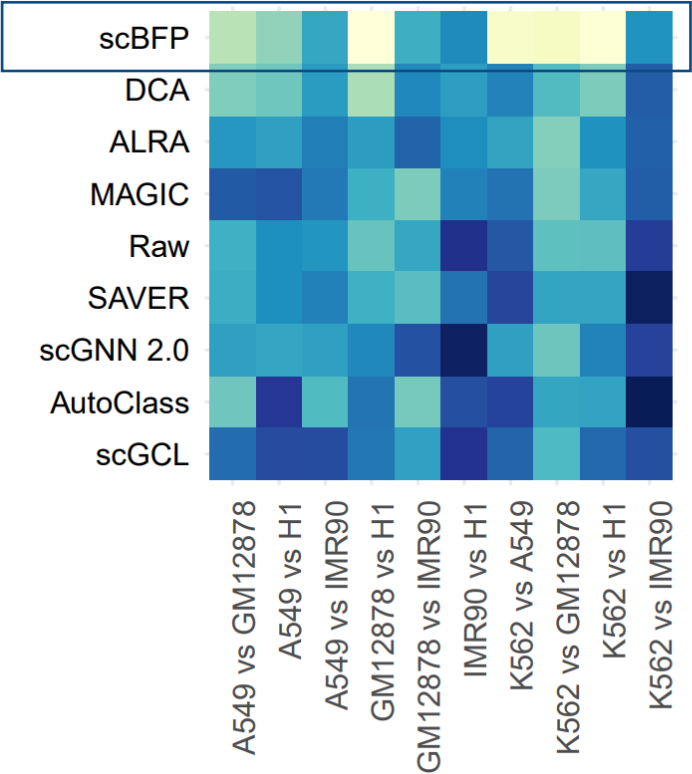
4 Experiments & Results

Marker Gene Expression Identification



Raw → scBFP expression levels of widely recognized marker genes

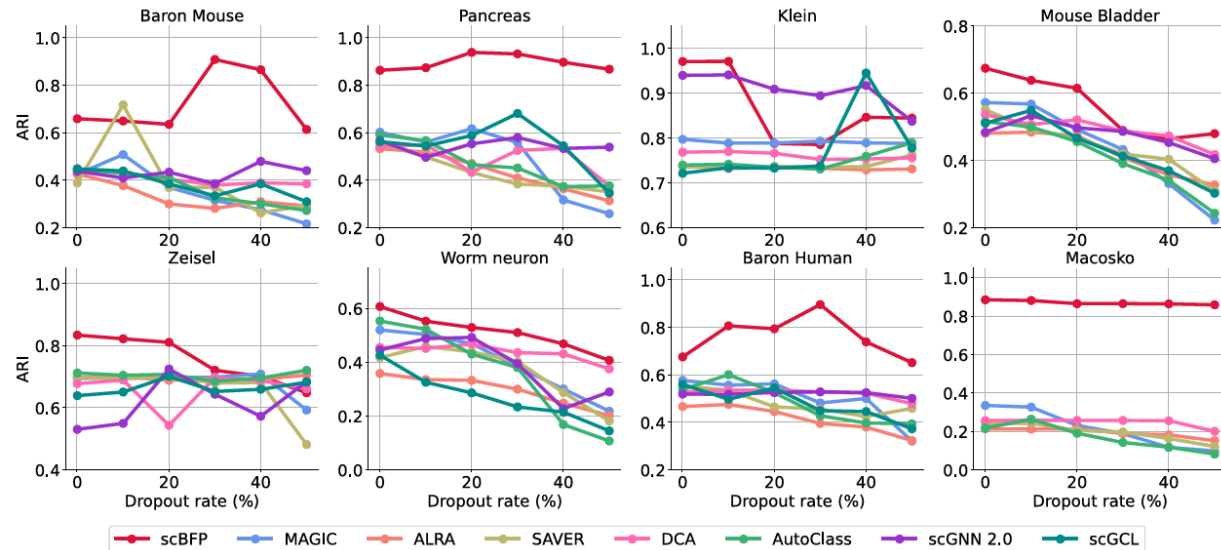
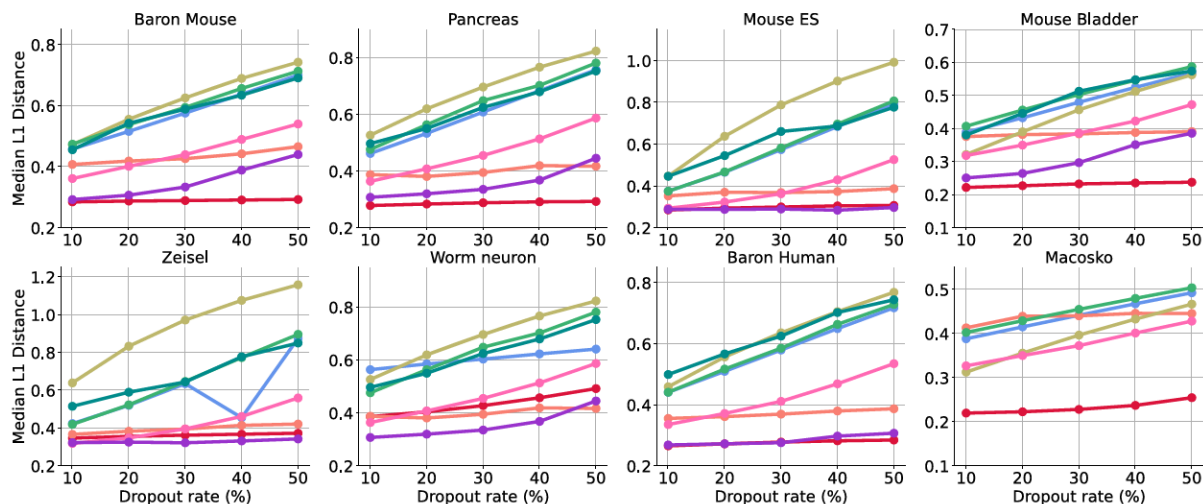
DEG Detection



scBFP achieves the highest DEG overlap with bulk RNA-seq references across cell-type comparisons, indicating improved recovery of biologically meaningful differential expression signals.

4 Experiments & Results

Dropout Recovery



By artificially masking non-zero values as dropouts, scBFP achieves stronger recovery with L1/RMSE and that this recovery translates into more robust clustering performance under increasing dropout rates.

Relevant Genes Enrichment in Lung Cancer Data

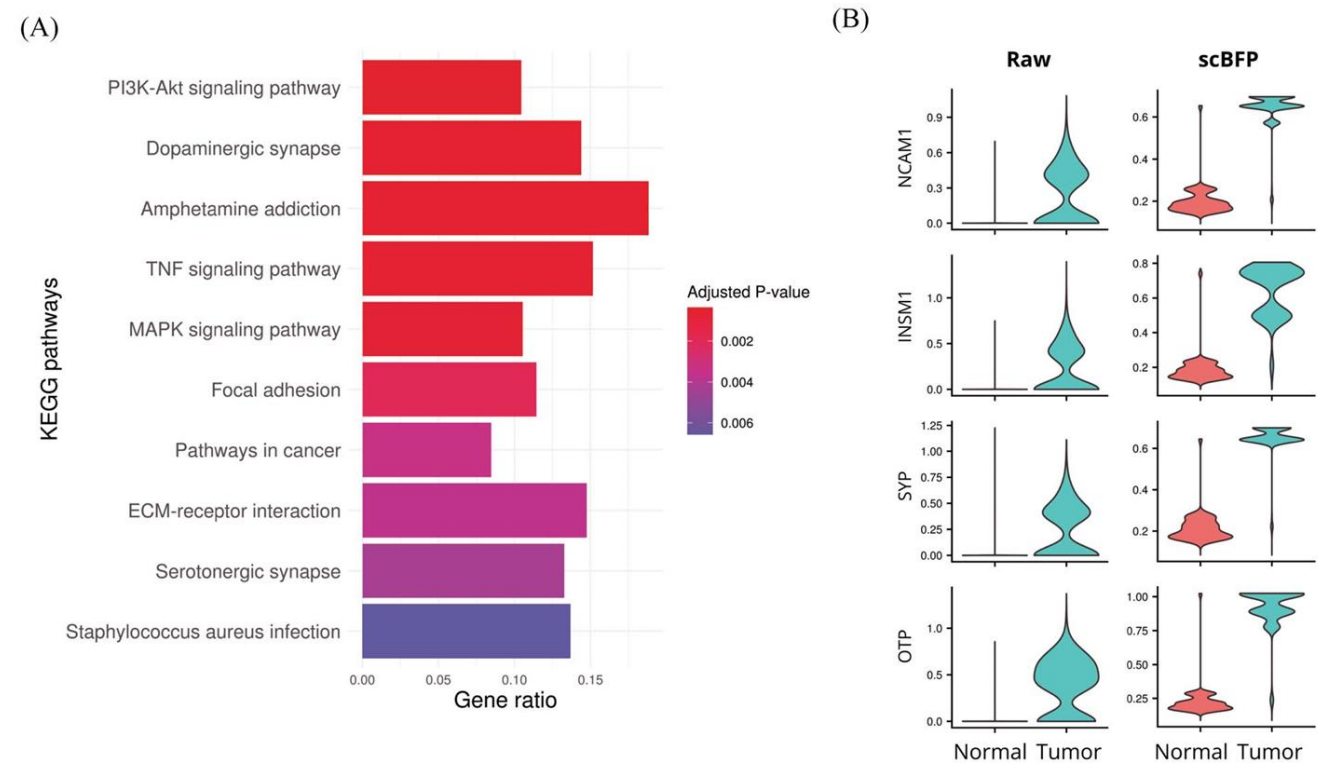


Figure 5. Differential gene analysis between lung carcinoid tumors normal lung tissues. (A) Uniquely enriched KEGG pathways from the data imputed by scBFP, compared to that from raw values. The top 10 pathways are displayed, ordered by their adjusted P-values. (B) Expression-level of scBFP with four markers for lung carcinoid: 'NCAM1', 'INSM1', 'SYP' and 'OTP'.

In lung cancer data, scBFP reveals biologically meaningful cancer-related genes and pathways more clearly, while enhancing the expression differences between tumor and normal tissues.

Scalability

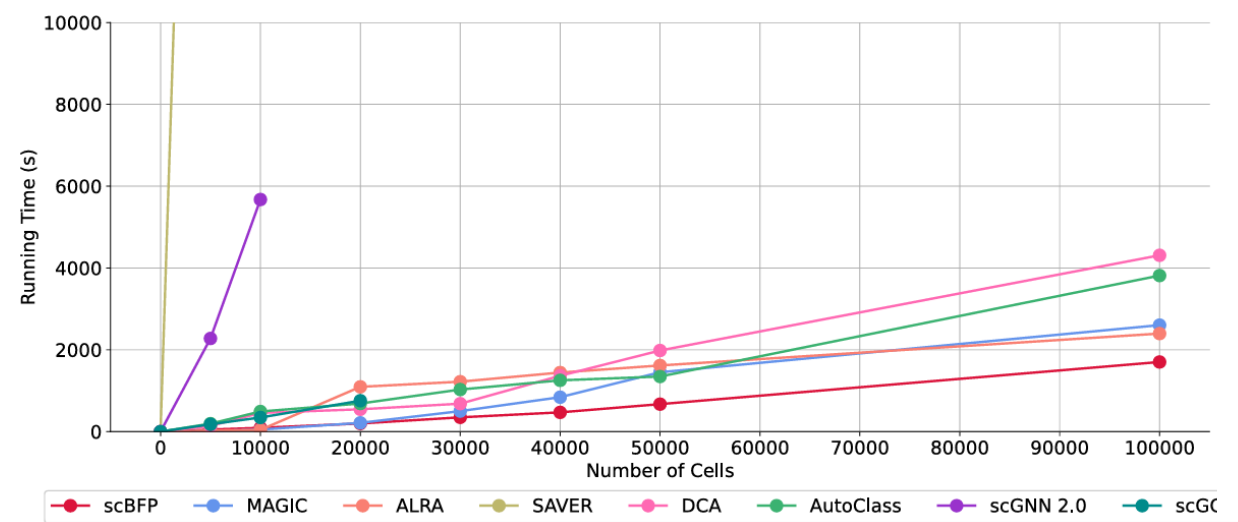


Figure 6. Running time comparison of scBFP and baselines across various number of cells with 10 000 genes.

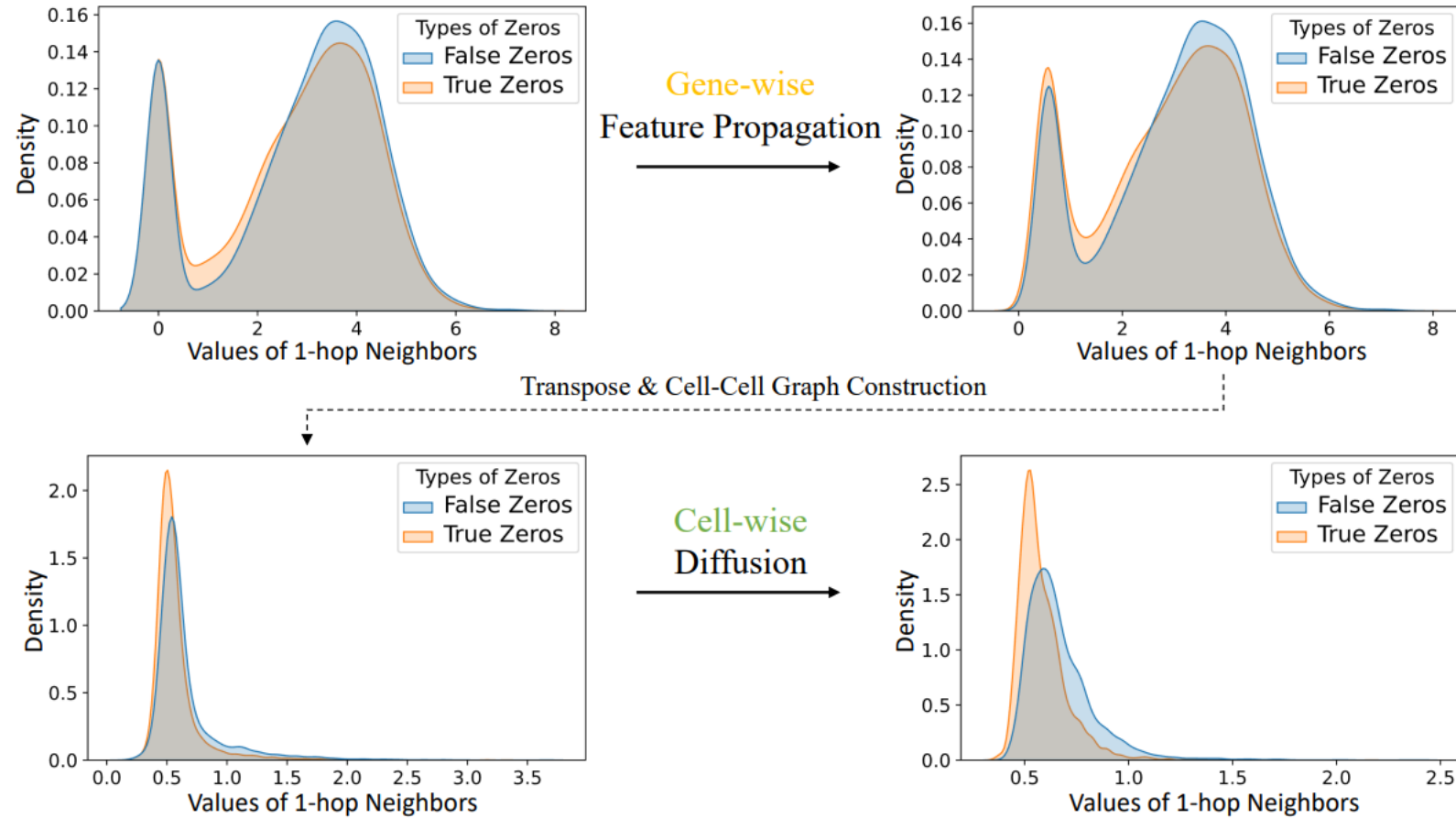
	Model input	Big-O
scBFP	$X, A^{\text{cell}}, A^{\text{gene}}$	$\mathcal{O}(BC) + \mathcal{O}(BG) + \mathcal{O}(\mathcal{E}_{\text{gene}}) + \mathcal{O}(\mathcal{E}_{\text{cell}})$
MAGIC	X, A^{cell}	$\mathcal{O}(GC) + \mathcal{O}(\mathcal{E}_{\text{cell}})$
ALRA	X	$\mathcal{O}(GC) + \mathcal{O}(Gk + k^2 + Ck)$
SAVER	X	$\mathcal{O}(GC) + \mathcal{O}(G) + \mathcal{O}(C)$
DCA	X, θ	$\mathcal{O}(GC) + \mathcal{O}(\theta)$
AutoClass	X, θ	$\mathcal{O}(GC) + \mathcal{O}(\theta)$
scGNN 2.0	$X, A^{\text{cell}}, \theta$	$\mathcal{O}(GC) + \mathcal{O}(\mathcal{E}_{\text{cell}}) + \mathcal{O}(\theta)$
scGCL	$X, A^{\text{cell}}, \theta$	$\mathcal{O}(GC) + \mathcal{O}(\mathcal{E}_{\text{cell}}) + \mathcal{O}(\theta)$

scBFP exhibits scalability, making it well-suited for processing high-throughput scRNA-seq data.

Key Summary

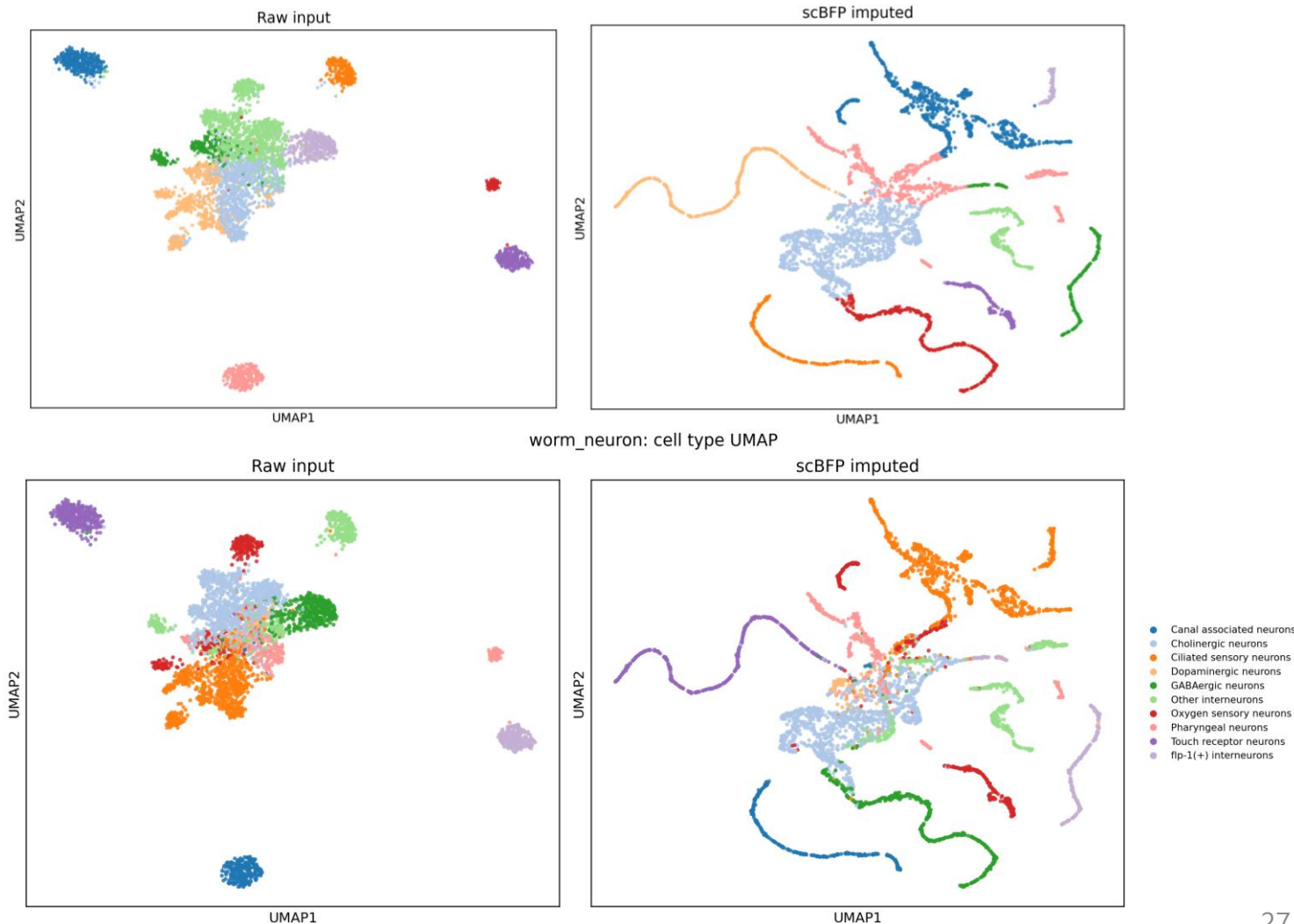
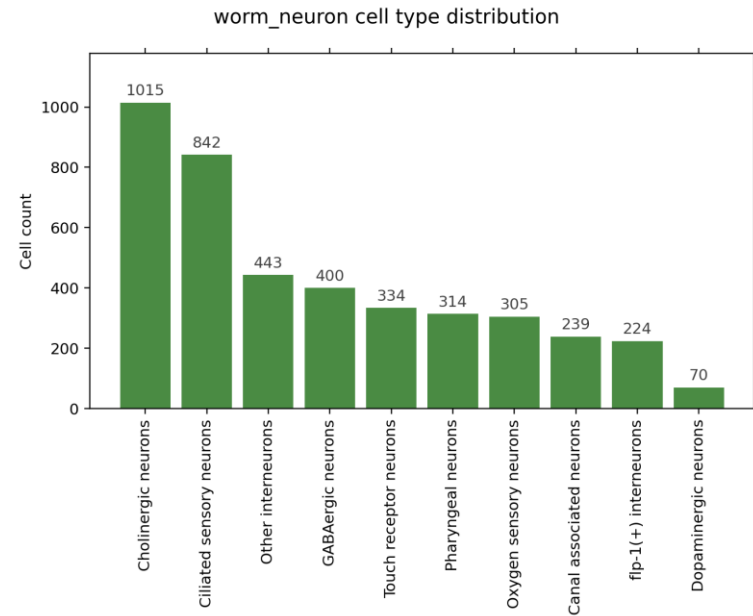
1. scBFP is a novel data imputation framework to mitigate false-zero propagation risk by using both gene-gene and cell-cell graph relationship information (enhanced graph structure).
2. scBFP generally shows improved performance on various downstream tasks(e.g., clustering, marker gene expression identification, DEG analysis, etc.) compared to the baseline methods.
3. scBFP demonstrates practical applicability (rare cell type identification and biologically meaningful insight extraction) and scalability for high-throughput scRNA-seq data.

More about True-Zero and False-Zero

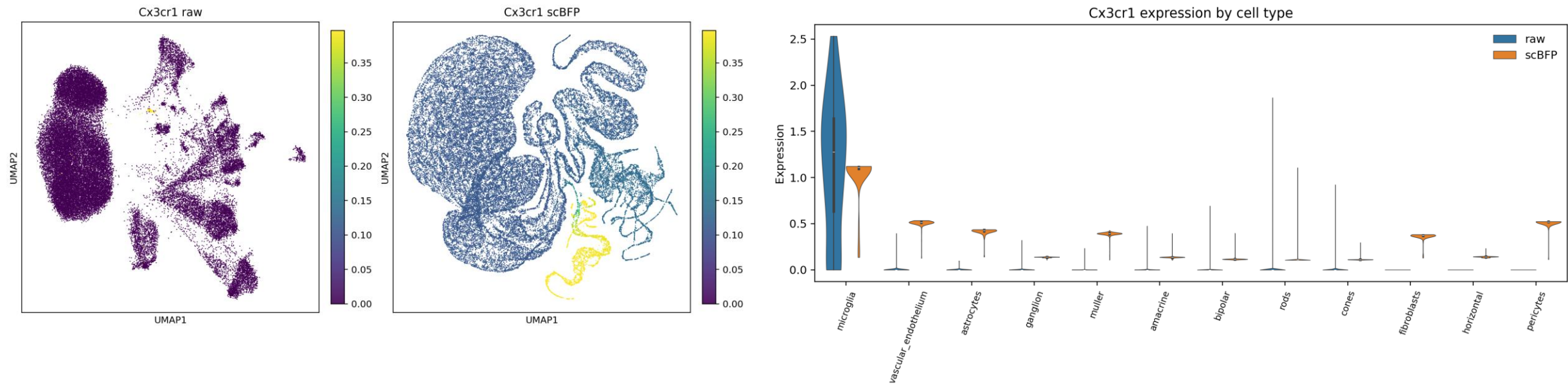


Clustering Performance Reproduction

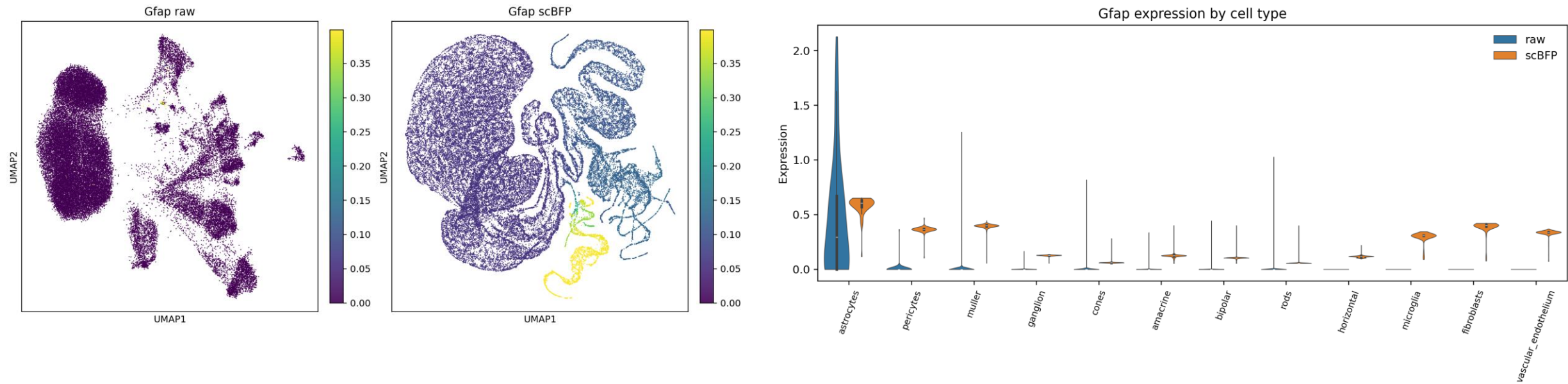
Dataset	Ari	Nmi	Ca
baron_human	0.68	0.83	0.73
baron_mouse	0.66	0.82	0.76
macosko	0.87	0.74	0.87
mouse_bladder	0.69	0.76	0.78
mouse_es	0.97	0.95	0.99
pancreas	0.86	0.9	0.92
worm_neuron	0.61	0.72	0.79
zeisel	0.83	0.79	0.87



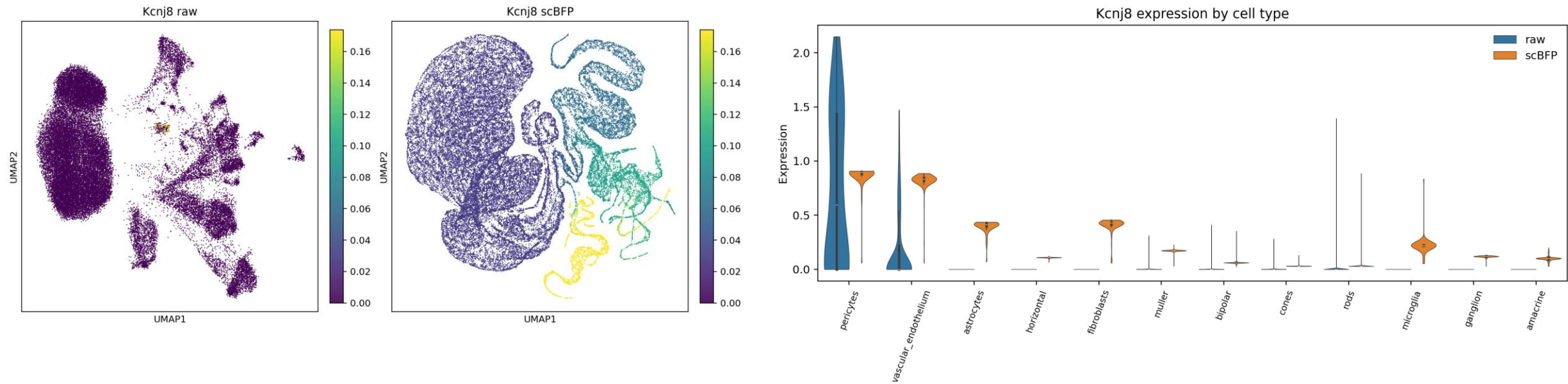
Rare Cell Type & Marker Gene Expression with Macosko Dataset



Rare Cell Type & Marker Gene Expression with Macosko Dataset

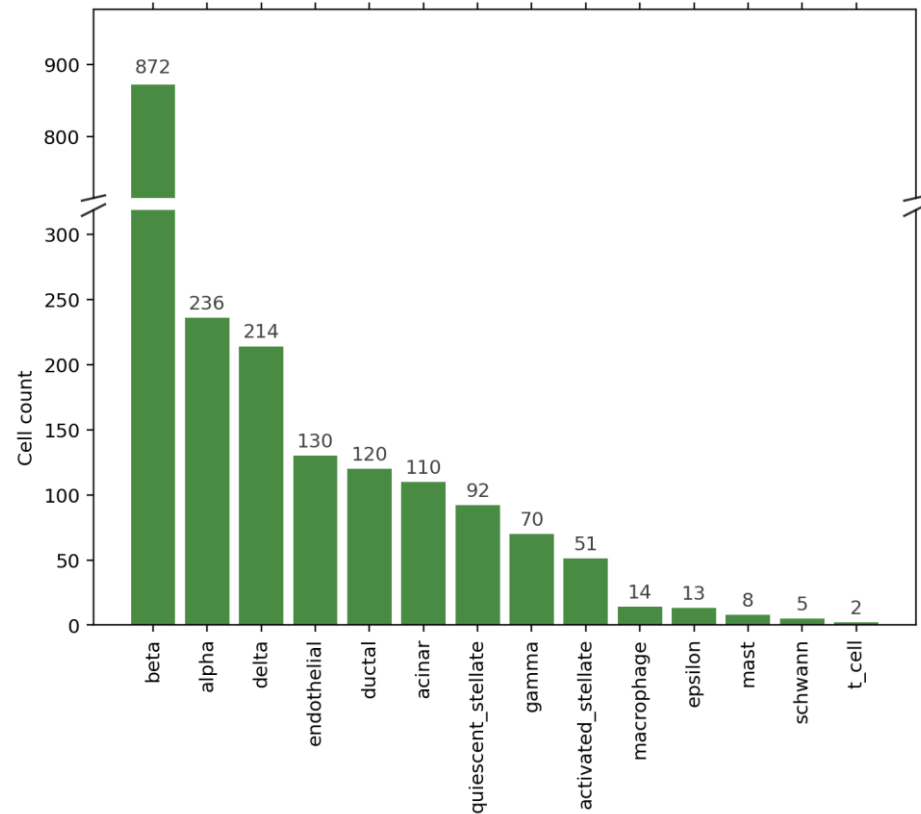


Rare Cell Type & Marker Gene Expression with Macosko Dataset

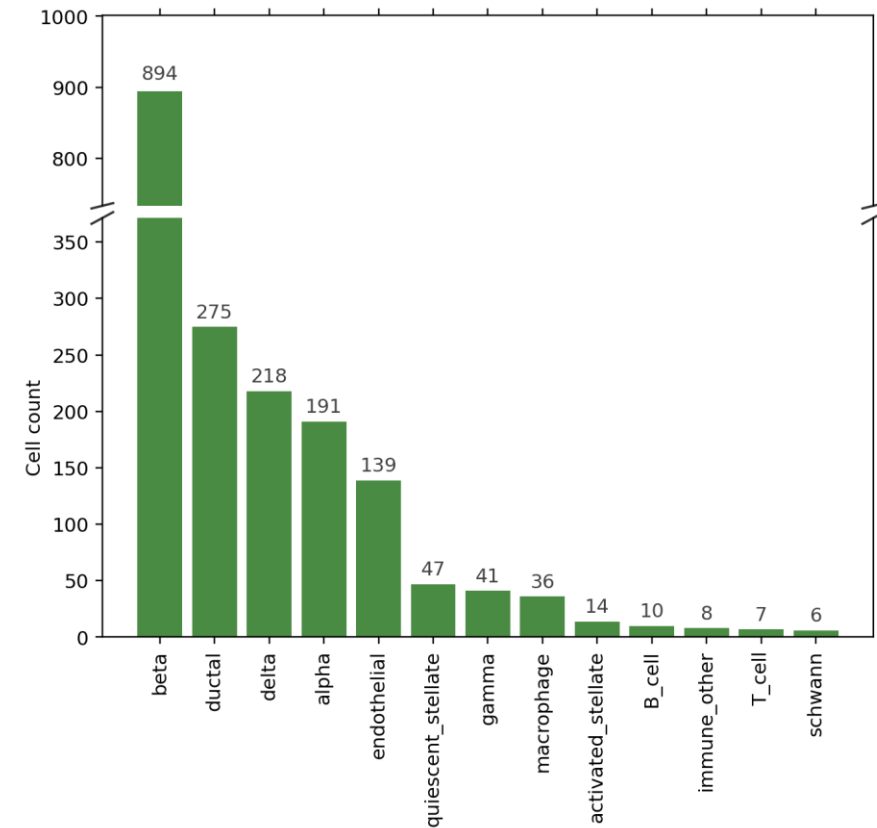


Extreme Rare Cell Type Case Study

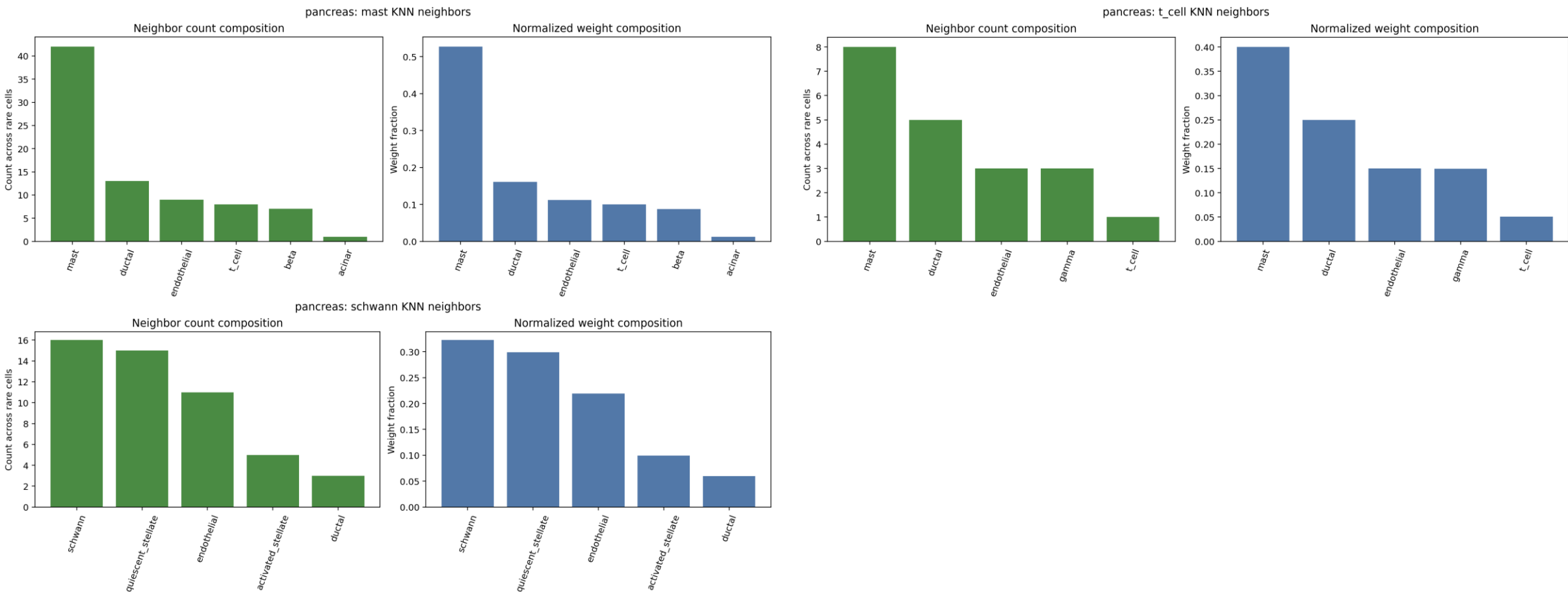
pancreas cell type distribution



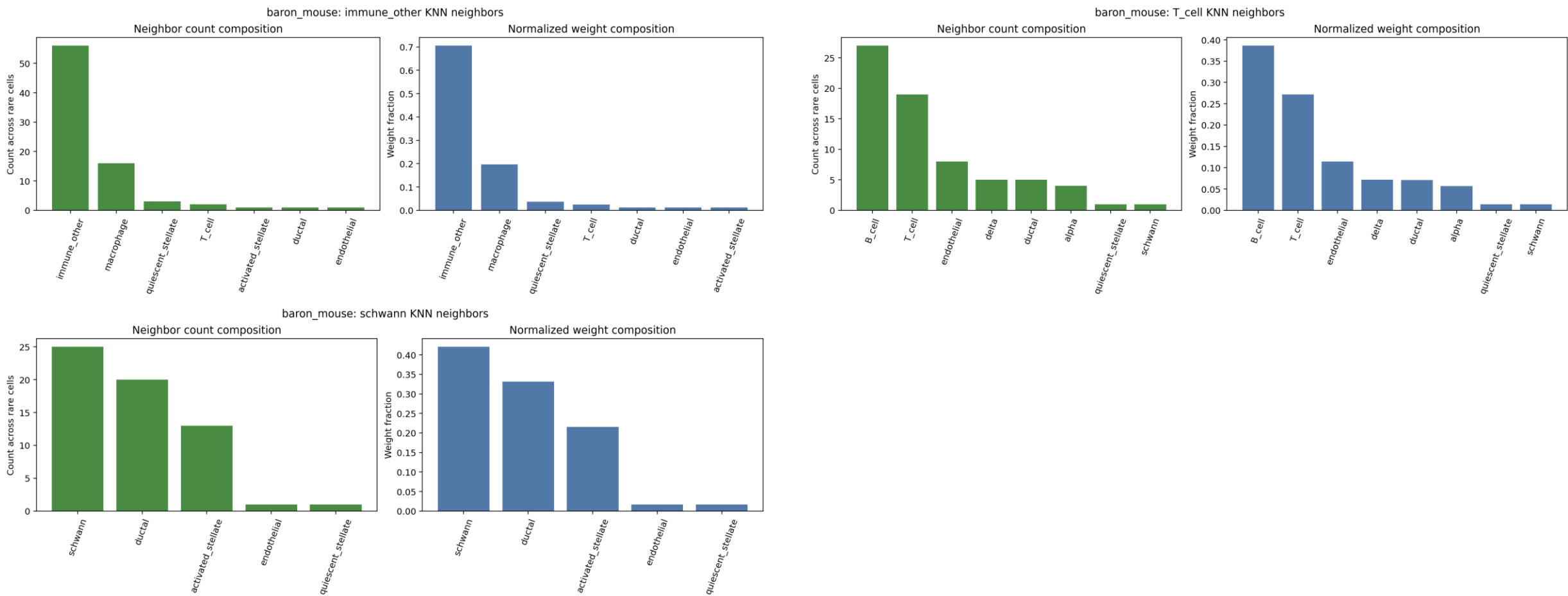
baron_mouse cell type distribution



Extreme Rare Cell Type Case Study



Extreme Rare Cell Type Case Study



Q&A